

A BT enterprise configuration management system

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Enterprise configuration management provides a framework within which to control the growth and complexity of modern large-scale integrated systems. BT is a leader in the field. This paper describes the company's design, implementation, and deployment of the Configuration Management Environment — a suite of whole-life, whole organisation, processes and tools needed to support a competitive and dynamic Networks and Systems division.

1. Introduction

The enterprise configuration management (ECM) concept and its relationship to traditional, project focused, configuration management (CM), is described elsewhere, together with many of the major elements of a practical implementation [1]. BT has implemented an ECM system and national deployment has commenced. This paper describes the processes and tools that, together, provide the BT Configuration Management Environment (CME).

2. The target enterprise CM environment

Figure 1 is a generalised view of the major physical and process elements targeted in early releases of CME.

Configuration items are the elements CME is attempting to control in terms of their parts, versions, and instances:

- parts are non-revisionable items within CME such as service platforms, operational support systems (OSS), system-interfaces, machines, and system-requirements — parts are logical, 'types of' entities used to define ownership, examples of such parts being the SMDS platform, CSS, or the Work Manager OSSs,
- versions are different configurations of items that go to make up a CME part (usually the manufactured instances of parts), examples of versions being SMDS V2.1, CSS Release 31, or Work Manager Version 5.0,
- instances are the deployed and, often replicated, configurations of versions, usually deployed for operational use but a reference or test environment is also a CME instance; examples of instances are the East Midlands CSS R31 or the Northern Home Counties (South) Work Manager V5.0.

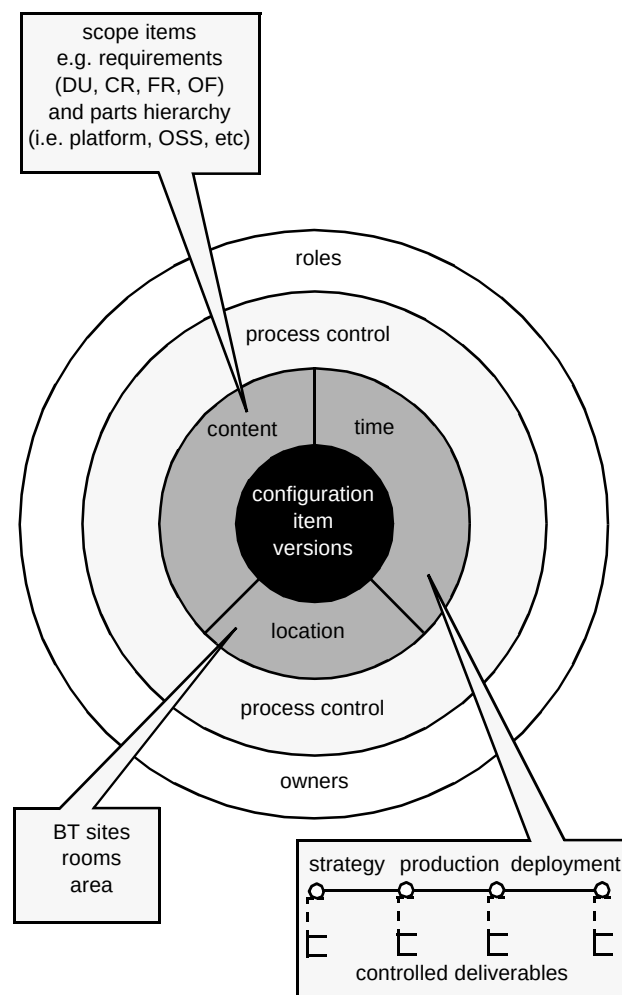


Fig 1 CME — the target environment.

Within CME, all of these configuration items are categorised according to their type. For example, CSS is a type of operational support system (OSS), SMDS is a type

of service platform, BT is an enterprise, and Tinsley Park computer centre is a type of location. These categories are at the root of a classification tree that is called the CME class hierarchy. Not all categories have been implemented in the early releases of CME but the complete list is shown in Table 1.

Table 1 The CME class hierarchy.

Category
Capability
Connection
Data
Enterprise
Equipment
Location
Network
OSS
Patch
Process
Programme
Requirement
Service platform
Software
Telecom product

In order to achieve the desired level of control within the enterprise-CM system, each configuration item is considered in terms of its content, time, and location, elements.

- Content identifies the reason for having the configuration item, the components that make up the item, and the relevant CME attributes of the item (e.g. specification, design, or test document references). CME is not a requirements management system nor a systems design tool; it maps and controls changes to content. Inadequate control of content leads to integration difficulties and unneeded or inadequate functionality.
- Time identifies the major process milestones and certain deliverables throughout the life of a configuration item. CME does not directly control the project plan; it maps and controls changes to milestone dates. Associated with each milestone there are various life cycle by-products known within CME as controlled deliverables. These are described later in this paper. Inadequate control of time leads to untimely delivery of content, investment, and *ad hoc* life cycle processes.
- Location identifies the configuration information about a deployed instance of a configuration item. Inadequate control of location leads to integration difficulties and costly, often embarrassing, delays in diagnosis and repair. CME is not an inventory system in the

conventional sense; it maps and controls configuration items to their physical, usually geographical, environment.

There is a further dimension not shown in Fig 1 — relationships. Relationships identify the connections and interactions between the components of a configuration item. Inadequate static and dynamic models, and shortfalls in control of relationships lead to unclear understanding of the interaction between content, time, and location. Table 2 shows the CME relationships and how they relate to relevant primary enterprise CM elements.

Table 2 CME relationships.

Relationship	Element
Compatible with	Content
Composed of	Content
Delivers	Content
Depends on	Content
In scope of	Content
Incompatible with	Content
Is benefited by	Content
Is targeted on	Content
Owns	Content
Replaces	Content
Uses	Content
Destination for	Content/Location
Source for	Content/Location
Has applied to it	Location
Has available	Location
Has deployed on it	Location
Located at	Location

Identification of configuration items at a level of abstraction consistent with enterprise-CM is not sufficient to ensure adequate control. Without a standard, flexible, process to monitor and control change throughout an item's entire life, the configuration would soon become outdated, inaccurate, and unreliable. Such a whole-life process is defined by CME and will be described later. It is sufficient to say here that each item type (e.g. platform, requirement, OSS, or location) has its own life cycle defined within CME.

The CME life cycle processes would not be of much value without the identification of major roles and the assignment of people, rather than groups or organisations, to those roles. Inadequate control of ownership leads to unclear responsibility and delays in resolving strategic, production or operational issues. In very large enterprises such as BT, standardisation of roles and assignment of the correct person to the role can be very difficult to track. Standard change-management roles are defined by CME and tracking of role assignment is carried out as the assignment changes with progression of a configuration item through the various stages of its life cycle.

The whole-life processes and change management control that is established by this target enterprise CM environment provide the framework within which to build solutions to the integration complexity problems that are mentioned earlier.

3. CME processes and information

There are two main process elements within an ECM system:

- standard life cycles that are appropriate to individual configuration items — these life cycle definitions must include stages, milestones, deliverables appropriate to each stage, and audit mechanisms that ensure completeness and accuracy at the beginning and end of each stage, even though all stages may not be used in a particular implementation,
- mechanisms by which changes to configurations are monitored and controlled — changes must be authorised by the right person at the right time without undue delay.

Much of the effort expended on CME to date has been concerned with the definition and implementation of these process elements because they do not exist outside individual production or deployment units. Where standard life cycles and change-control processes do exist, they focus on traditional, project CM items such as specifications, source code, or design documents.

3.1 Life cycles

Over many years, there has been much discussion and controversy over the most appropriate life cycle for any

particular production process. This discussion will no doubt continue as organisations and technologies change.

It is not the task of CME to solve the life cycle dilemma or to introduce yet more controversy, but to define the essential elements of a lifetime process that facilitate monitoring and control of configuration items. With this in mind, the life cycles assigned to each CME item have been designed to map to the way people currently do their job and are flexible enough to integrate with existing, lower level, more detailed processes.

Figure 2 shows a typical CME life cycle. There are really two life cycles shown in the diagram:

- a top-level life cycle that has strategy (solution design and evolution planning), production (systems engineering), and deployment (operations) phases, which is known as the CME whole-life life cycle,
- a more detailed life cycle having nine stages, called the CME generic life cycle, to which all item life cycles map.

Most people working within a whole-life life cycle phase will be familiar with the detail of that part of the life cycle. For example, most production people are aware of define, acquire, integrate, system test, and customer acceptance test phases; most deployment people are aware of deployment and in-service phases. However, many people engaged on production or deployment are not aware of the strategy phase.

This lack of awareness of other major parts of the whole-life life cycle leads to erratic quality in the handover from one phase to the next. There is also a lack of

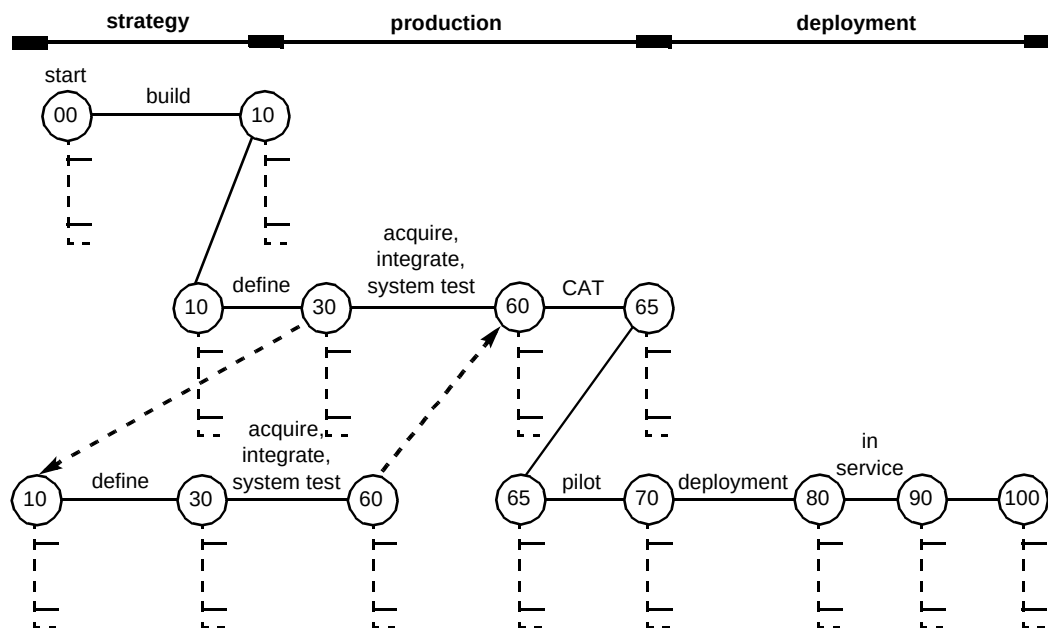


Fig 2 The CME generic lifecycle.

understanding and definition as products or services move from inception through production to operational use and eventual withdrawal.

A good example of this lack of co-ordination is withdrawal. Many BT operational support systems have a lifetime of several years, or even decades, and gradually become redundant as their functionality is replaced by one or more, newer systems. At some point their operational costs outweigh their usefulness but the people responsible for strategy and production have moved on to other things and the original reason for having the system may be lost in the mists of time. The system consumes valuable human and computing resources but, as there is no trigger to initiate the withdrawal process, these resources are not freed for other uses. How many software licences are paid for annually and never actually used?

CME ensures that its configuration items are tracked and audited throughout their entire lifetime, not just a single, parochial, phase. Tracking a product or service along a single life cycle is often not sufficient. In the case of large integrated solutions, for example, several major system components may have to co-operate to provide the required level of functionality and this means there may be multiple life cycles running in parallel but asynchronously. These life cycles must be coupled if the integration is to succeed efficiently. In the example of Fig 2 this is shown by the subordinate life cycle starting at the 'acquire' phase and ending at the 'integrate and test' phase. Re-synchronisation occurs at quality gates (see section 5).

Delivery time-scales of major products and services are often extended because there is insufficient notice of changes to the delivery time-scales of subordinate components that lie on the critical path. At the same time, other subordinate components are produced before they are really needed. Investment capital and skilled resources are tied up unnecessarily.

CME's visibility of the complete, integrated set of life cycle dependencies and its ability to monitor milestone dates ensures that change and its impact can be assessed against business priorities before it is too late to invoke risk avoidance strategies.

The introduction of business logic to future releases of CME will help to increase business efficiency by making better use of process and data information already visible to the system.

There are two other features of CME life cycles shown in Fig 2, namely:

- the circles with the numbers inside are the life cycle milestones, or quality gates, controlled within CME, numbers being used to avoid the ambiguity associated

with textual phrases such as 'the gate at the end of the acquisition phase' — also, experience during the pilot trial of CME shows that people are more at ease with 'doing a 10 gate',

- the 'tails' growing out of the quality gates indicate the list of controlled deliverables appropriate at that milestone.

Quality gates and controlled deliverables are discussed in more detail in the next two sections.

3.2 Quality gates

As configuration items progress down their respective life cycles, there are significant events when decisions must be made about their suitability to progress to the next phase. These events are the major milestones defined and controlled within CME and the associated process is the quality gate process.

Quality gates are not new to most commercial engineering organisations and are an integral part of BT's delivery and quality management systems. The novelty introduced by CME is that a standard suite of gates and a simple, supported process has been defined for all CME configuration item types within BT's Networks and Systems (N&S) division.

Figure 3 shows a portion of a life cycle with a focus on the production phase. This is based on the example described in section 3.1 above. At the gate 30 milestone, the OSS should have a clear, accurate, top-level, definition of requirements and design to be passed to subordinate external or internal suppliers. However, experience shows that without an adequate, standardised process and a clearly defined set of essential phase deliverables, configuration items often pass through this important milestone incomplete, inaccurate, and without clear role or ownership assignment.

The standard CME quality gate process entails taking a baseline by the transmitting authority and acceptance or rejection of the configuration item by the receiving authority. The decision to accept or reject the item is supported by the certificate (this is a CME report showing, among other things, the delivery status of certain phase deliverables, their current owners, their new owners, and their storage location). At this point, the configuration is baselined on CME, thus providing a record of a known, accepted or rejected, configuration status. Modifications to the baseline may also be recorded. A decision to progress to the next phase must be supported, on CME, by a request to change the life cycle status of the configuration item.

Future planned milestone dates will also be confirmed or modified as part of this gate process and, if required,

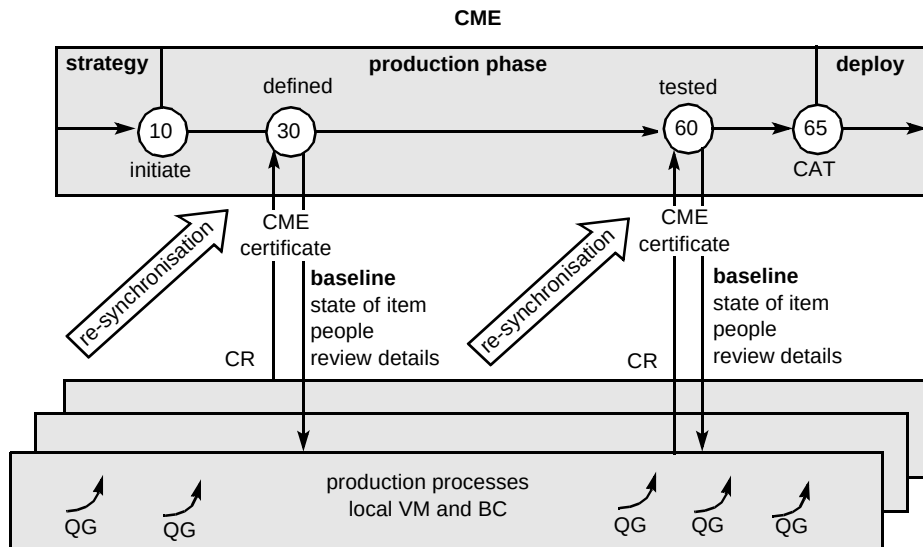


Fig 3 CME quality gates.

changes will be controlled within CME. This way, the impact of early or late deliveries may be assessed at each quality gate. All of this activity, for all CME controlled items, is visible across all N&S departments by all people who have suitable access to CME. Of course, as not everyone will want to see all details for all items, filtering mechanisms are required (see section 4.1).

3.3 Controlled deliverables

Some configuration items start their lives in the strategy phase, driven by market or regulatory requirements and pass through all of their CME defined life cycle phases. Others never pass beyond the inception phase. Others are externally supplied, complete or subordinate components with a variety of lifetimes and major milestones. Whatever the life of a configuration item, the process through which it passes gives rise to certain by-products some of which may be regarded as essential to progress and others merely desirable.

In any event, it is an important element of the CME control process that the essential by-products are adequately monitored with regard to quality, status, and timeliness of delivery. Within CME the, usually essential, by-products are predefined according to each phase of each configuration item life cycle. They are known as 'controlled deliverables' and these provide the basis for assessment of progress, completeness, and accuracy at quality gates. Table 3 shows a sample of controlled deliverables required throughout the life cycle of an OSS and an OSS instance.

It is important to note that CME neither stores these deliverables nor tracks the detailed, underlying production, review, and configuration management processes. This responsibility lies with the creators and owners of the deliverables. CME maintains some status information about

controlled deliverables, and pointers to libraries of compatible documents, throughout the item life cycle. An imminent release of CME will provide on-line, drill-down access from CME to certain types of deliverable and lower level storage repository.

Table 3 Sample controlled deliverables.

OSS	OSS instance
Process milestone dates Build list document CAT results CM plan document Declared deficiencies Deployment design document Design spec document Pilot report Scope statement User guide document IVV&T strategy Installation guide Test summary document	BIS date INFOMAN CR No. Installation date Operational status Switch settings Decommission date Source identifier

3.4 Change control framework

So far, not much has been said much about how CME will implement or maintain control over those configuration items, their attributes (e.g. milestone dates and controlled deliverables), or their relationships, for which it is responsible. The core component of the control mechanism, and hence much of the power of the ECM system, is the CME change control framework.

With so many different life cycles, item types, and organisations involved in the whole-life, whole-N&S approach to CM, it is essential that the CME change control mechanisms are relatively simple to operate. They must also be rigorous in their definition and flexible in their

application. Otherwise, frustrated users and wasteful delays will ultimately cause degradation of CME information and defeat the purpose for which it is intended.

Much of the design and implementation effort expended in CME production has been associated with change control. Proprietary configuration management tools were evaluated at the outset of the CME project. None of these provided, or readily facilitated by simple reconfiguration, the level of process control, role assignment, and class inheritance needed to support the comprehensive suite of ECM principles and processes needed by BT.

The following principles, established during the preliminary design stages of CME, provide the framework for change control process definition and implementation.

- Each CME item type has its own life cycle — a major distinction between traditional, project focused CM and enterprise level CM is the number and variety of the life cycles needed to support their process objectives. There are approximately two dozen item types defined by CME, from assembly, through enterprises, interfaces, interface instances, and peripherals, to telecom products. Each of these has a life cycle with two to nine major milestones or quality gates. Obviously, lower level CM environments will have more detailed life cycles and milestone requirements. This is a matter for the local CM authority.
- Item life cycles map on to the generic life cycle. CME maintains a whole-life view of configuration status by integrating strategy, production, and deployment life cycle components. Previously, these were perceived to exist in isolation while, in fact, they were closely linked and dependent. This perception of isolation introduces a lack of understanding about the needs and difficulties experienced in other generic life cycle phases. Consequently, the opportunity does not exist for early, team-like, resolution of problems across the whole life of an item. The result is often late delivery, inability to operate, or under or over specified functionality, of strategic BT products and services. By mapping to a single life cycle, CME introduces a common language of control.
- Generic roles and owners must be identified. Without clear, unambiguous, ownership, by real people rather than groups or organisations, there is a shortfall when it comes to identification of authority, responsibility, and accountability. Without a set of standard roles, there is no objective way of viewing ownership. CME defines a small, but important, set of generic roles (e.g. Change authority, change requestor, interested party, and change executioner) to which people may be assigned. These people may also have operational roles such as evolution planner, programme manager, integration manager, operations manager, or team leader but part of the CME modelling exercise, described in section 5.1 below, is to map operational roles to CME roles. CME insists on item ownership even if it has to trace the owner back through the classification hierarchy to the default owner, a CME administrator.
- Detailed change is masked by abstraction. Another major distinction between enterprise-CM and tradition CM is that the former does not attempt to maintain control of every item and activity across the enterprise. Such a process would be too inflexible, require technologies that have not yet been standardised nor implemented, and cause considerable commercial upheaval. Local processes and people filter detailed change before relevant information is logged with CME. The linkage between the layers of CM is mainly procedural with abstraction being achieved through the configuration item's primary attributes only. Accuracy is enforced by the quality gate process.
- Baselines are taken at important milestones. This provides an agreed record of the configuration item's content, time, and location elements at discrete points in time. In theory, it should be possible to roll back to this baseline configuration. In practice, a complete roll-back is unlikely to occur because of component size, the complexity of inter-component dependencies, and the close relationship between item-status and time. Partial roll-back is more likely where non-delivery of one configuration item (e.g. a requirement) will facilitate on-time or strategic delivery of the remaining package.
- Change notification and alerts are automatic. The previously mentioned CME principles need standard process and deliverable definitions to be successful, this principle needs a change control sub-process based, mainly, on cognitive perception and an acute understanding of the operational use within the enterprise. Otherwise, an unsolicited stream of unwanted change requests will bombard many people. Again, this has been one of the most difficult parts of CME to implement and test because there is not a readily available proprietary product.
- The delivery mechanism, e-mail or other network solutions, is commonly available. The ability to distinguish between those known to need the change information, those known and suspected to need the information, and those who might be interested in the information is not available at all.
- The current release of CME implements a design of automatic alert/notification, implied by role, that will prove its technical and algorithmic viability in the longer term.

In certain circumstances standard CME processes may be inappropriate. If a full quality gate assessment is avoided or a controlled deliverable not delivered, an exception change request must be submitted to CME so that the event is logged and the authorisation verified.

3.5 Summary

If the ECM problems detailed in Moor et al [1] are to be resolved, the description of CME above should provide a logical solution by now. Table 4 shows a simple, general mapping from problem to CME process as described above.

Table 4 Problems versus solutions.

Problem [1]	Solution
Scale, complexity and variety	Life cycles, quality gates
Visibility of the big picture	Integrate life cycles
Management of change	Change control framework
Management of reuse	Categories, availability
Definition of whole-life deliverables	Controlled deliverables
Audit processes	On-line reporting
Disjoint life cycle phases	Whole-life views

When the process, information and supporting tool-set is implemented and deployed, CME will not only overcome current difficulties but also have the ability to provide a rapid response to future regulatory, commercial, and technological requirements.

The next few sections of this paper describe how CME has been designed, implemented, and deployed.

The Integration Handbook [2] has already been aligned to the CME life cycles and deliverables. It has also been distributed to many people involved in the CME generic life cycle, especially Computer Support and Operations (CSO) divisions. As it evolves, the Integration Handbook coverage will extend and it will underpin CME in its deployment right across the life cycles.

A small team, drawn from across BT's Networks and Systems (N&S), has defined item categories, class hierarchy, life cycles, quality gates and process standards. The team could have defined this ECM system to be more sophisticated and detailed but instead has designed it to be easily accessible by users, flexible, and non-dictatorial. CME is not about mandating quality standards; it is more about mandating risk assessment.

4. CME design

Modern distributed object technology has had a considerable influence on the design of CME. Distributed object technology is particularly well suited for creating flexible client/server systems because the data and business logic are encapsulated within objects, allowing them to be located anywhere within the distributed system. Distributed objects have the inherent potential to allow granular components of software to interoperate across networks, run on different platforms and coexist with legacy systems through object wrappers.

In the large, complex, highly integrated world of telecommunications, this design approach is the only way to provide an enterprise-CM solution. The solution must be flexible and allow sufficient scope for evolution as the enterprise goes through changes to its organisation, products and services portfolio, and systems portfolio.

Another design decision taken at the beginning of the project was that, where possible, CME components must be bought not made. As things turned out it has not been possible to acquire as many proprietary components as one would like because they are simply not available. This introduced unforeseen difficulties when implementing CME but most of the issues have now been resolved.

4.1 Architecture

Figure 4 provides a brief overview of the CME three-tier architectural model.

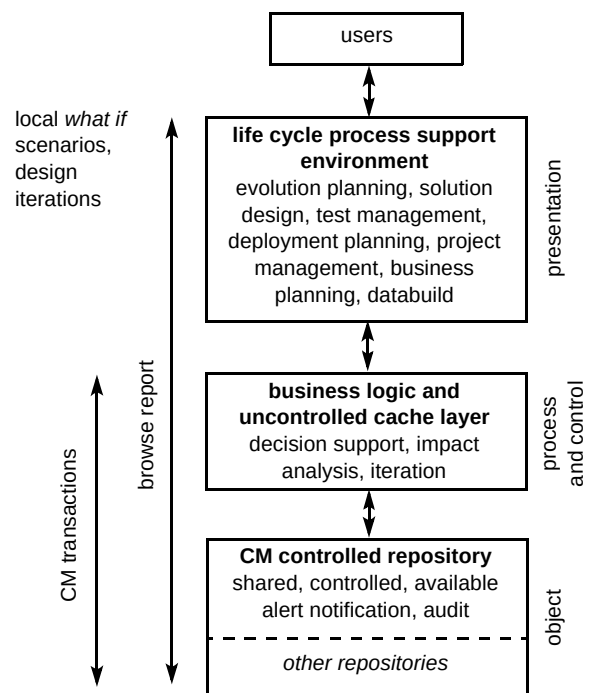


Fig 4 The CME three-tier model.

The object layer encapsulates the storage, metadata (interfaces and attributes), and basic operations (e.g. create item, delete item, alert notification and auditing). This layer hides a shared interface to other valuable CM repositories, such as the national capability register (NCR), software engineering centre (SEC) local project CM repositories, the major deliverables database (MDDb), the project finance system (IRIS), and software distribution tools (e.g. XFER).

The process and control layer encapsulates the business logic at the enterprise level. The object layer handles short-lived process functions. The process and control layer is really a specialisation of the object layer that handles long-lived processes and the ECM environment at large. For example, it knows how to handle a workflow or long-lived transaction in furtherance of decision support or impact analysis operations.

The presentation layer displays information, rising from object and process & control layers, visually to the user. There are a variety of presentations available depending on the user-type (e.g. senior manager, evolution planner, deployment planning, or databuild).

The presentations may communicate directly with the object layer or through the process and control layer. The Telecommunications Requirements, Impact Analysis, and Decision Support (TRIADS) and millennium compliance systems are examples of this layer.

All three layers co-operate to provide a single-object view, with the underlying division of labour transparent to the user or other distributed objects.

4.2 Physical design

Figure 5 provides a top-level view of the current CME physical design. A Digital Equipment Company (DEC) machine acts as host to World Wide Web (WWW), configuration management tool, e-mail, and IVA servers.

Netscape, as the BT preferred tool, provides WWW browser access. The CM tool used is PCMS produced by SQL Software Ltd (Hertford) and, again, this is a BT preferred tool. The IVA server is a small in-house built piece of software that provides a transparent link between the CME browser and the PCMS tool.

CME browser access is provided by a small client implemented using the Visual Basic programming language. A Netscape client provides WWW browser access. Various e-mail clients are used depending on the user's environment.

In general, CME browser access and alerts are provided over the BT Multi-protocol Router Network (MPRN) while CME change management is performed over the WWW.

The initial data capture activity, carried out at the beginning of each CME deployment, uses a Microsoft Access database.

Operator access and CME administration functions use X-windows or text terminals according to the CME component being accessed.

The highly distributed nature of CME together with its need to co-operate with other new and legacy BT systems dictated the requirement to have a standard, open, object-based infrastructure. CME uses the Common Object

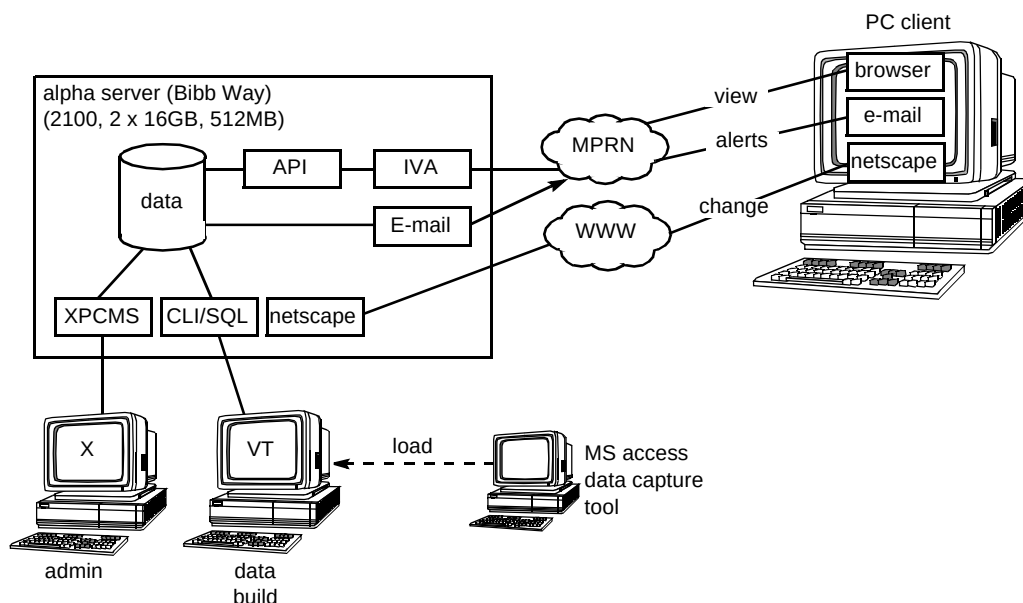


Fig 5 CME physical design.

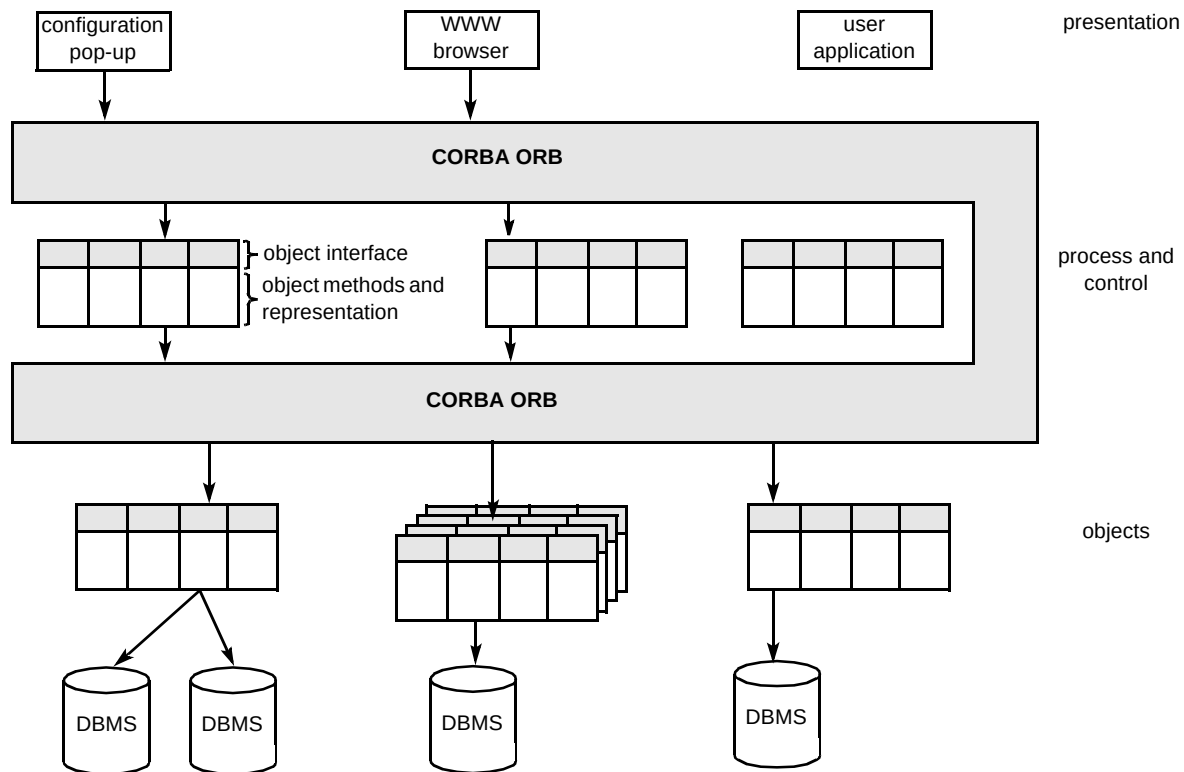


Fig 6 The CME CORBA bus.

Request Broker Architecture (CORBA) presented by the Object Management Group (OMG).

Figure 6 shows how CME uses CORBA to implement the three-tier architecture described in section 4.1 above.

5. CME deployment

5.1 The process

Figure 7 shows the process initiated when an OSS, assembly, or service platform commences deployment on CME.

An introductory session describes the deployment process, and a demonstration of the CME functionality, user interface, and data collection tools sets the foundation for later, more practical sessions.

A number of interactive workshops follow the introductory session and during these sessions relevant people, usually those who will maintain data or operate CME in their own environment, produce a model of parts, items, instances, and relationships. This model is captured using the two Microsoft Access database tools, PDATA and IDATA, that hold part and item data respectively. The model is then automatically transferred to a temporary CME environment, known as dataload where the model is checked for consistency and correctness.

Another set of interactive workshops then takes place, during which the model is refined and made ready for general use within the target project or operational support domain. During this phase, prospective users are also trained on the use of the CME change control processes and tools. CME data is moved to the support environment in readiness for trial operation.

Figure 8 shows a typical CME model. Construction of these models for large integrated systems is difficult and time consuming, especially if system design or support and operation information is weak. The modelling process is something of a paradox; if accurate information were readily available, BT would probably not need the same CME functionality anyway. Experience during pilot trials and actual CME deployment has shown that the company needs CME.

Interfaces are a particularly important aspect of all integrated systems and modelling these can be quite a daunting task. Figure 9 shows a typical CME interface model.

An interesting, and very important, issue that has caused considerable frustration during large-scale integration, is the question of interface ownership. Responsibility for technical design is usually resolved by assigning authority to the transmitting or receiving system but this still leaves open the question — 'Who owns the interface?' The answer lies in the application of the platform framework [1] and the CME class hierarchy. Interfaces are owned by the next level up in the component's structure. Hence, in the example

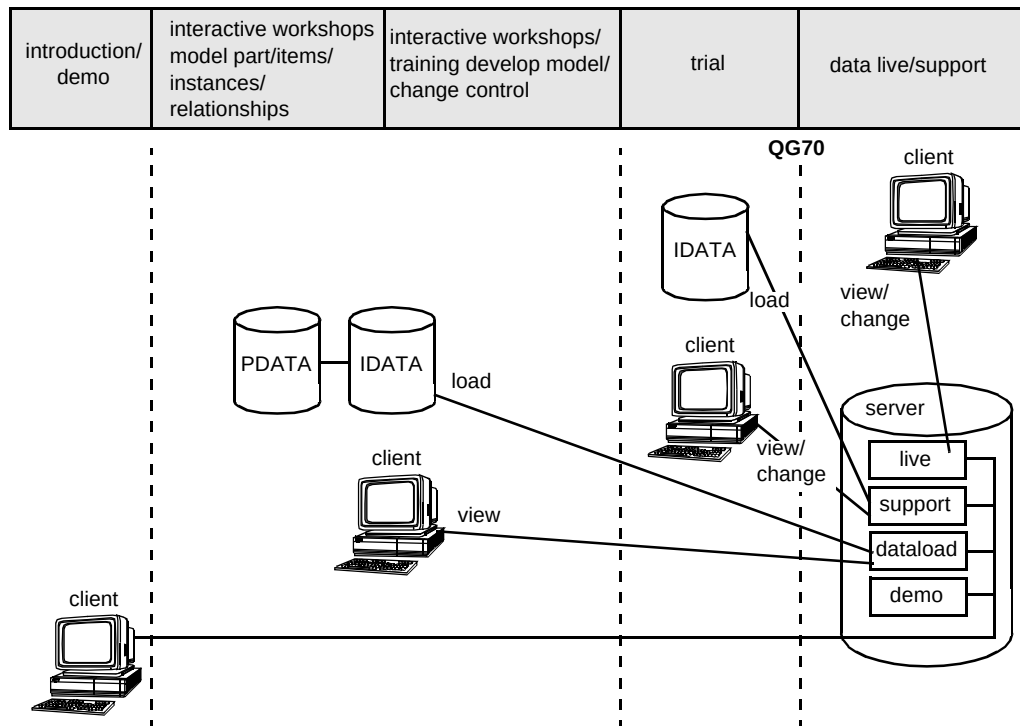


Fig 7 The CME deployment process.

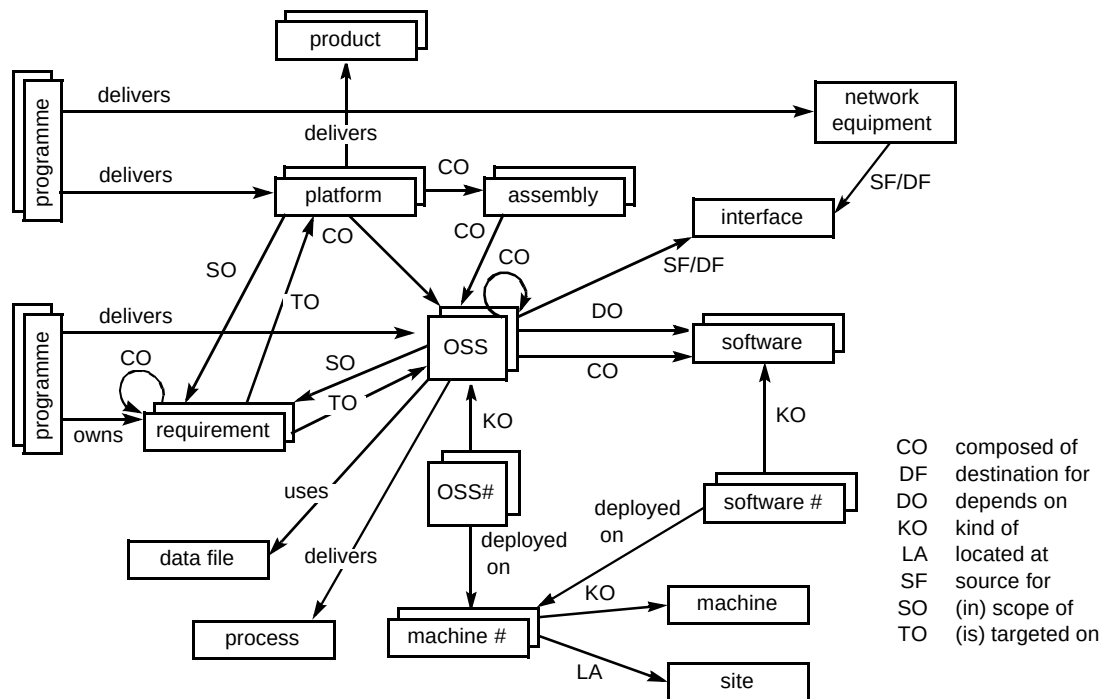


Fig 8 A typical CME model.

shown in Fig 9, the interface between OSS_A and OSS_B is owned by the ASSEMBLY_A, the interface between PLATFORM_B and PLATFORM_C is owned by ENTERPRISE_C, and so on.

to delivery, diagnosis, and repair of products and services in a large, integrated, commercial IT environment.

When the interactive workshops, training, and model building is satisfactorily completed, there is a short trial period when the users, assisted by the CME deployment team, have an opportunity to operate CME as though it were

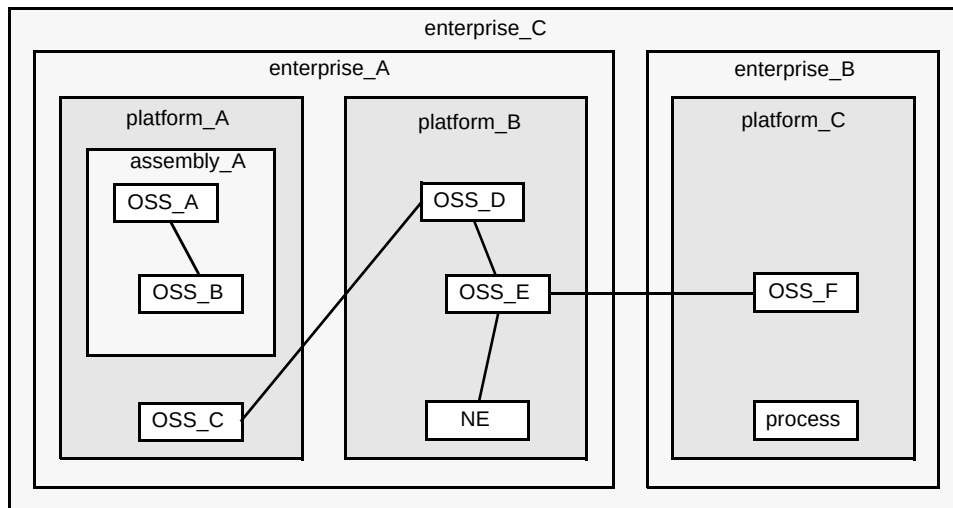


Fig 9 A CME interface model.

a live environment. At the end of this period (CME quality gate 70), a go-live date is decided.

Before live operation commences, the CME deployment team moves the CME data to the live environment where it is integrated with the operational enterprise model.

The amount of CME and project effort and the elapsed time required when undertaking a particular deployment of CME depends on the size and complexity of the model required. This in turn depends on the level of control needed by the target enterprise component.

5.2 The deployment schedule

It has taken longer than expected to establish a trained deployment team but this is now in place and CME deployment has started. A one-year detailed plan is agreed with the ECM programme and CME project manager. Approximately one hundred target domains are scheduled for inclusion in CME by the end of 1997. Figure 10 shows the position at the end of February 1997.

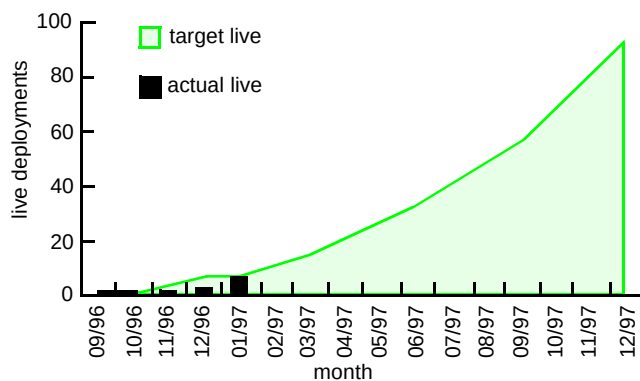


Fig 10 CME deployment on-load.

Table 5 shows a list of target domains that have either commenced deployment or have achieved live status on CME. The list includes some of BT's largest and most strategic service platforms and operational support systems.

Table 5 Active or live deployments.

BTSE (Data)	OMC
CME	OMS/CBP
CSS	PDH Platform
ESMF/OSMF	Platform Framework
Featurenet Platform	Platforms Evolution
FMS	PSTN/ISDN Platform
FRACT	SDH Platform
IT infrastructure	ServiceView
JV (set meal)	SMART
MCSS	Systems Evolution
Millennium (data)	Systems Framework
NCR	VBS
NCR (data)	Work Manager

The first few deployments are the most difficult because no one wants to be first even if they are eager to gain the CME benefits. However, it becomes progressively easier to take on new target domains as the quantity of CME data and size of the enterprise model increases.

In addition, as CME becomes an integral part of everyday business-as-usual activity, trained people will instinctively train others.

A high priority objective of the ECM programme is to complete full national deployment of BT's ECM system within three years. During this period, the major share of CME project budgets will be expended on deployment rather than management or implementation activities.

6. ECM research contracts

A BT research contract with the CM Research Foundation, Leeds University, shows that BT is a leader in the field of ECM. Although substantial benefits of the existing CME have already been identified, quantified and presented to N&S senior management, if BT is to maintain its lead and increase ECM benefits, further research work is needed.

A brief review of current proprietary and research material produces no available solutions to important ECM issues associated with the following:

- enterprise data/process quantity and variety,
- automated process management and control using integrated business logic,
- cost-effective ways to quickly reconfigure existing enterprise components.

The BT Corporate Research Programme has accepted an outline proposal for research work in principle. This work aims to produce novel approaches in three topic areas:

- visualisation of data complexity,
- modelling of component capability,
- benchmarking/modelling of integrated business logic.

7. Conclusions

CME provides a framework within which all of the problems defined at the beginning of this paper can be resolved. This is achieved mainly by hiding complexity, definition of standard processes, and by providing visibility of change throughout a whole, unified life cycle.

The BT implementation of an enterprise configuration management system (CME) has taken approximately two years from concept to the initial stages of full-scale, national deployment.

More recently, as BT's merger partner, MCI, has shown a keen interest in ECM and CME in particular, modelling products and services across enterprises may become a reality in the near future.

CME is more than just another configuration management tool. It has a major role to play in decision

support, impact analysis, risk analysis, and process re-engineering. For the first time, one can drill down from an individual product or service, through the service platform, to the networks and management systems on which the product or service is supported.

Potentially, one could track a BT product to individual source code files in a software engineering centre, or a memory chip description in a CSO inventory, or a solution design or floor layout in other diverse repositories.

With a fully populated CME, just-in-time production and deployment of major, large-scale, integrated systems become a real possibility.

Finally, to answer a frequently asked question, CME is modelled and controlled using CME.

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Steve Moor graduated from the University of Sussex in 1974 with a degree in Electronic Engineering and joined BT to take up a post in transmission planning and works. He joined Systems Development in 1979 to work on System X. In 1984 he joined Network Management Department (NMD). In 1992, after several years working on OMC and TNS in development and project management roles, he took up a post to establish a single point of systems design within NMD.

Two years later, when the design unit was established and consistent processes and methods to address solution design as well as system and deployment design had been adopted, he moved to Systems Integration department. He became the integration manager for INET. During this time the need for enterprise CM within BT became more critical. He has been managing the ICM project since 1995.



Keith Gleen joined BT as an apprentice in 1962. Following a period in London, he moved to the research department, where, in the early 1970s he worked on 'Telecommunications in the 1990s' — which aimed at delivering integrated and broadband services to the home. He then moved into computing and on to System X, packet switching and a variety of jobs involved with data communications. In 1990, he moved to the network management area where he became involved in the evolution and establishment of the platform framework. He is now innovation manager in Networks and

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